LAB REPORT: DATA DEFINITION LANGUAGE (DDL)

CSBB 204: Database Management Systems

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Aim

To understand and execute Data Definition Language (DDL) commands: `CREATE`, `ALTER`, `RENAME`, `TRUNCATE`, `DROP`, and constraint operations.

Theory

Command	Description
`CREATE`	Defines a new database object such as a table, index, or view.
`ALTER`	Modifies the structure of an existing object (add/drop/modify columns and constraints).
`RENAME`	Changes the name of an existing database object.
`TRUNCATE`	Removes all rows from a table without logging individual row deletions; cannot be rolled back in most DBMSs.
`DROP`	Deletes an entire object from the database schema.
`DELETE`	DML command that removes selected rows; can be rolled back and fires triggers.

SQL Solutions and Output

(a) Create `employee_details` and rename to `emp_details`

```sql
CREATE TABLE employee_details (
    eid       INT PRIMARY KEY,
    ename     VARCHAR(50),
    eaddress  VARCHAR(100),
    dob       DATE
);
```

```sql
ALTER TABLE employee_details RENAME TO emp_details;
```

Verification:

```sql
SHOW TABLES;
```

Tables_in_aryan
department
dependent
dept_locations
emp_details
employee
project
works_on

(b) Remove all records from `emp_details`

```sql
TRUNCATE TABLE emp_details;
```

(c) Add column `contact_no VARCHAR(10)`

```
ALTER TABLE emp_details ADD contact_no VARCHAR(10);
```

Verification:

```
DESCRIBE emp_details;
```

Field	Type	Null	Key	Default	Extra
eid	int	NO	PRI	NULL	
ename	varchar(50)	YES		NULL	
eaddress	varchar(100)	YES		NULL	
dob	date	YES		NULL	
contact_no	varchar(10)	YES		NULL	

(d) Modify column to accept non-null values only

```
ALTER TABLE emp_details MODIFY ename VARCHAR(50) NOT NULL;
```

Verification:

```
DESCRIBE emp_details;
```

Field	Type	Null	Key	Default	Extra
eid	int	NO	PRI	NULL	
ename	varchar(50)	NO		NULL	
eaddress	varchar(100)	YES		NULL	
dob	date	YES		NULL	
contact_no	varchar(10)	YES		NULL	

(e) Drop column `contact_no`

```
ALTER TABLE emp_details DROP COLUMN contact_no;
```

(f) Add and drop table constraints

```
ALTER TABLE emp_details ADD CONSTRAINT uq_ename UNIQUE (ename);
```

```
ALTER TABLE emp_details DROP CONSTRAINT uq_ename;
```

(g) Drop `emp_details` using CASCADE and RESTRICT
Using CASCADE:

```
DROP TABLE emp_details CASCADE;
```

Using RESTRICT:

```
DROP TABLE emp_details RESTRICT;
```

Difference:

- **CASCADE:** Automatically drops all dependent objects (views, foreign key constraints, triggers) along with the table. No error is raised even if dependencies exist.
- **RESTRICT:** Refuses to drop the table if any dependent objects exist. The operation fails with an error, protecting against accidental data loss.

(h) Difference between `TRUNCATE` and `DELETE`

Aspect	TRUNCATE	DELETE
Command Type	DDL	DML
WHERE Clause	Not supported (removes all rows)	Supported (selective removal)
Transaction	Auto-commit; cannot rollback	Can be rolled back
Performance	Faster (deallocates pages)	Slower (logs each row)
Triggers	Does not fire triggers	Fires DELETE triggers
Identity Column	Resets AUTO_INCREMENT	Preserves counter value
Locking	Table-level lock	Row-level locks

(i) Create an index on the Employee table

```
CREATE INDEX idx_emp_ename ON employee(ename);
```

Verification:

```
SHOW INDEX FROM employee;
```

Table	Non_unique	Key_name	Seq_in_index	Column_name	Index_type
employee	0	PRIMARY	1	eid	BTREE
employee	1	idx_emp_ename	1	ename	BTREE

(j) Add `email VARCHAR(30)` to the Dependent table

```
ALTER TABLE dependent ADD email VARCHAR(30);
```

Verification:

```
DESCRIBE dependent;
```

Field	Type	Null	Key	Default	Extra
eid	int	NO	PRI	NULL	
dependent_name	varchar(50)	NO	PRI	NULL	
relationship	varchar(20)	YES		NULL	
email	varchar(30)	YES		NULL	

(k) Create `product_catalog` table

```
CREATE TABLE product_catalog (
    prod_id     INT,
    prod_name   VARCHAR(50),
    prod_price  DECIMAL(10, 2),
    stock_qty   INT
);
```

(l) Add `category VARCHAR(20)` to `product_catalog`

```
ALTER TABLE product_catalog ADD category VARCHAR(20);
```

(m) Set `prod_id` as the primary key

```
ALTER TABLE product_catalog ADD CONSTRAINT pk_prod_id PRIMARY KEY (prod_id);
```

(n) CHECK constraint: `prod_price > 0`

```
ALTER TABLE product_catalog
ADD CONSTRAINT chk_price_positive CHECK (prod_price > 0);
```

(o) Make `stock_qty` NOT NULL

```
ALTER TABLE product_catalog MODIFY stock_qty INT NOT NULL;
```

(p) Add `expiry_date` of type DATE

```
ALTER TABLE product_catalog ADD expiry_date DATE;
```

(q) Default value of 0 for `stock_qty`

```
ALTER TABLE product_catalog MODIFY stock_qty INT DEFAULT 0 NOT NULL;
```

(r) Add `discount_rate DECIMAL(5,2)`

```
ALTER TABLE product_catalog ADD discount_rate DECIMAL(5, 2);
```

(s) Default value of 10 for `discount_rate`

```
ALTER TABLE product_catalog MODIFY discount_rate DECIMAL(5, 2) DEFAULT 10;
```

(t) CHECK constraint: `discount_rate` between 0 and 100

```
ALTER TABLE product_catalog
ADD CONSTRAINT chk_discount_range CHECK (discount_rate BETWEEN 0 AND 100);
```

Final structure:

```
DESCRIBE product_catalog;
```

Field	Type	Null	Key	Default	Extra
prod_id	int	NO	PRI	NULL	
prod_name	varchar(50)	YES		NULL	
prod_price	decimal(10,2)	YES		NULL	
stock_qty	int	NO		0	
category	varchar(20)	YES		NULL	
expiry_date	date	YES		NULL	
discount_rate	decimal(5,2)	YES		10.00	

(u) Drop `product_catalog` using RESTRICT

```
DROP TABLE product_catalog RESTRICT;
```

(v) Drop `product_catalog` using CASCADE

Note: Table must be recreated before executing CASCADE drop.

```
DROP TABLE product_catalog CASCADE;
```